ATM Menu-Driven Banking Simulation

1. Problem Statement

Develop a Python program to simulate ATM operations such as deposit, withdrawal, balance inquiry, and transaction summary with proper validation and account limits.

2. Objectives

- Deposit money.
- Withdraw money.
- Check account balance.
- Display transaction summary.
- Validate transactions and account limits.

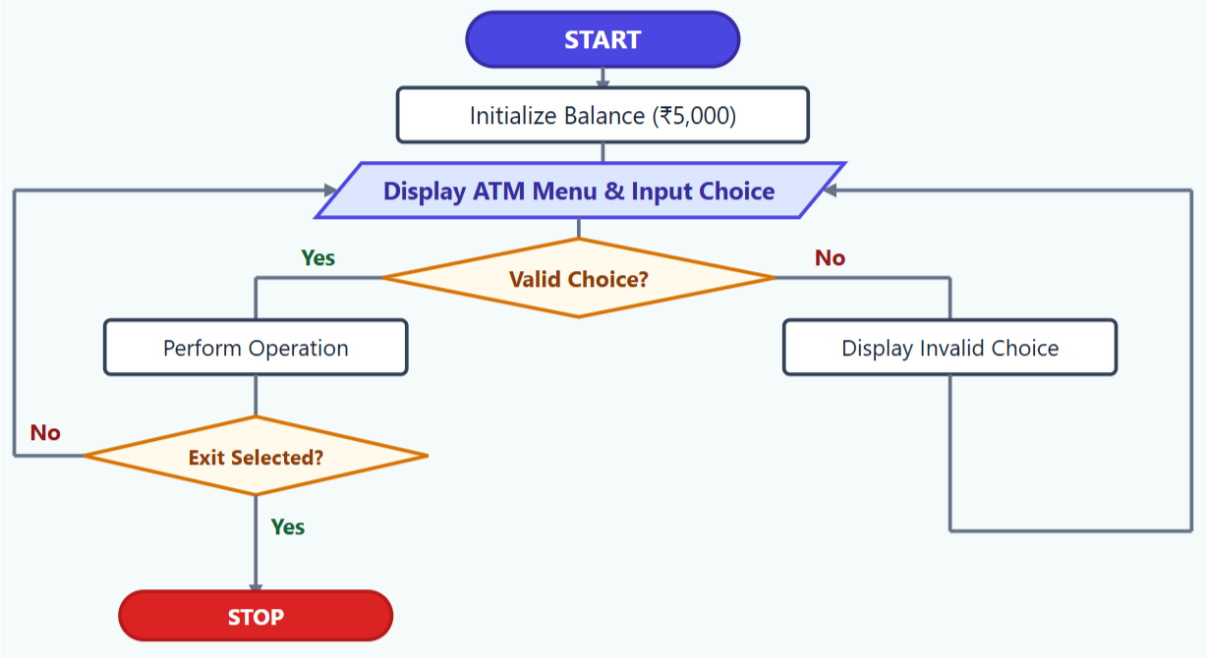
3. Assumptions

- Initial balance: ₹5,000.
- Maximum withdrawal: ₹10,000 per transaction.
- Minimum balance: ₹500.
- Deposit limit: ₹20,000 per transaction.
- Invalid amounts are not accepted.

4. Algorithm

1. Start.
2. Initialize account balance.
3. Display ATM menu.
4. Select an operation.
5. Perform deposit, withdrawal, or balance inquiry.
6. Validate transaction limits.
7. Display transaction summary when selected.
8. Repeat until Exit.
9. Stop.

5. Flowchart



None

6. Pseudocode

None

```
START

balance = 5000

REPEAT:
    Display menu
    Read choice

    IF choice = 1:
        Deposit money

    ELSE IF choice = 2:
        Withdraw money

    ELSE IF choice = 3:
        Display balance
```

```
    ELSE IF choice = 4:
        Display transactions

    ELSE IF choice = 5:
        EXIT

    ELSE:
        Display invalid choice

STOP
```

8. Sample Output

None

```
1. Deposit
2. Withdraw
3. Balance Inquiry
4. Transaction Summary
5. Exit

Enter choice: 1
Enter deposit: 2000
Deposit Successful

Enter choice: 2
Enter withdrawal: 1000
Withdrawal Successful

Enter choice: 3
Balance: Rs. 6000.0

Enter choice: 4
Transaction Summary:
Deposit: Rs.2000.0
Withdrawal: Rs.1000.0

Enter choice: 5
Thank You!
```

9. Test Cases

Test	Operation	Expected Result
1	Deposit ₹2,000	Success
2	Withdraw ₹1,000	Success
3	Withdraw ₹15,000	Invalid
4	Deposit ₹25,000	Invalid
5	Balance inquiry	Display balance
6	Exit	Stop program

10. Conclusion

The ATM simulation performs deposits, withdrawals, balance inquiries, and transaction summaries using Python loops, conditions, lists, and input validation.